Analytics & Big Data

Session 4: Regression I

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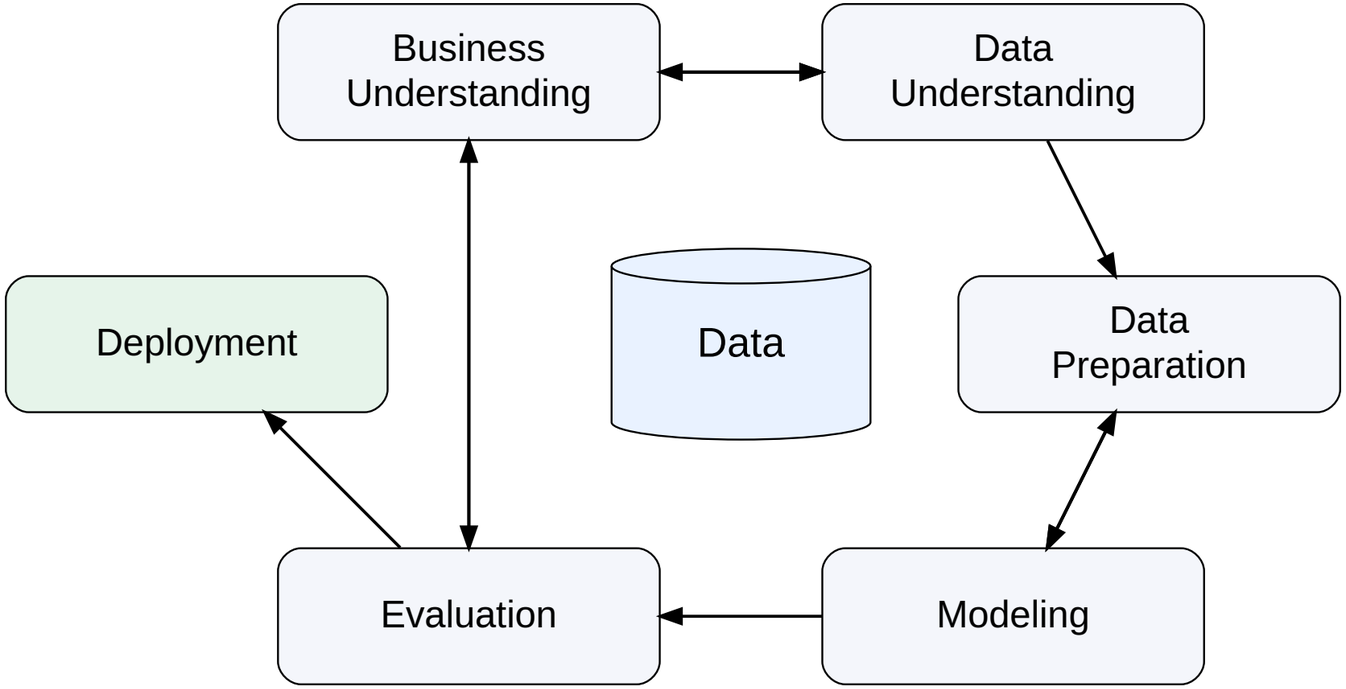
🎯 **Learning objectives**

- **Explain** the stages of a model-based analytics workflow using linear regression as an example.
- **Interpret** linear regression models, including coefficients, OLS estimation, and model evaluation.
- **Describe** how regression models are implemented in Python.

Process

In this session, we follow the CRISP-DM process:



Business understanding

Case: House prices — What drives property value?

Housing markets represent one of the largest asset classes in most economies. Residential real estate accounts for a substantial share of household wealth, and even small pricing errors can translate into large financial consequences.

Understanding what drives house prices is therefore relevant for:

- **Real estate firms** use pricing models to advise clients.
- **Banks** rely on valuation models to assess collateral and manage mortgage risk.
- **Insurers and policymakers** use property data for risk assessment, taxation, and urban planning.

How could we use analytical models, such as regression models, to understand the drivers of prices?

Data understanding

Dataset: Ames Housing (Kaggle)

To answer our question, we need a dataset that includes:

- Sale prices
- Physical attributes (e.g., size, rooms)
- Location characteristics
- Quality indicators

To address this, we turn to a **publicly available dataset** on Kaggle: the Ames Housing dataset, which provides detailed information on residential properties and their sale prices.

About Kaggle

Kaggle is a popular data science platform offering:

- Public datasets
- Notebooks for analysis
- Competitions
- An active community

Load the data

As a first step, we retrieve the dataset and load it into our Python environment for analysis.

```python
import pandas as pd

USAhousing = pd.read_csv('../exercises/data/ames.csv')
USAhousing.head()
```

	Order	PID	area	price	MS.SubClass	MS.Zoning	Lot.Frontage	Lot.Area	Street	Alley	...	Screen.Porch	Po
0	1	526301100	1656	215000	20	RL	141.0	31770	Pave	NaN	...	0	0
1	2	526350040	896	105000	20	RH	80.0	11622	Pave	NaN	...	120	0
2	3	526351010	1329	172000	20	RL	81.0	14267	Pave	NaN	...	0	0
3	4	526353030	2110	244000	20	RL	93.0	11160	Pave	NaN	...	0	0
4	5	527105010	1629	189900	60	RL	74.0	13830	Pave	NaN	...	0	0

5 rows × 82 columns

> An important step is to **understand the meaning of the variables**—that is, what each column represents and how the data was collected.

Understanding the variables

To understand the variables, we consult the **dataset documentation** and create a structured overview.

We create a small table that summarizes key variables:

Variable	Meaning	Unit / Values	Notes
`Order`	Unique order number	—	Check for duplicates, NA not allowed
`PID`	Parcel identification number	—	Same property can appear multiple times (e.g., repeated sales)
`area`	Above-ground living area	Square feet	Check for **plausible ranges and consistency**
`price`	Sale price of the property (target variable)	USD	Check **format, missing values, and outliers**
`MS.SubClass`	Type of dwelling	Categorical codes (e.g., 020, 060)	Retrieve **code definitions** and check consistency with other variables
`MS.Zoning`	General zoning classification	Categorical (e.g., RL, RM, FV)	TODO: Understand categories (may require **external expertise**)
`...`	Additional variables (e.g., lot size, street type)	—	…

> this step often involves **acquiring access to data, extracting it from systems, consulting documentation, talking to domain experts, and making sense of how the data was collected and defined**.

For the Ames Housing dataset, we rely on this data description: http://jse.amstat.org/v19n3/decock/DataDocumentation.txt

Data preparation

Prepare and explore the data

Before estimating a regression model, we first **prepare and explore the data**. Key steps include:

- **Check data quality**
  - What are the units and possible values?
  - Missing values, duplicates, inconsistencies
  - Plausibility of values (e.g., extreme prices or sizes)
- **Format and transform variables**
  - Numeric vs. categorical variables
  - Encoding categories, scaling if needed
- **Explore relationships**
  - Summary statistics
  - Distributions and scatter plots

> **Note**
>
> These steps were covered in detail in the **data preparation lecture**. Here, we focus on the **analytical modeling**.

Modeling

Model choice

1. Specify prediction task

- Define the **target variable** (e.g., price as a continuous variable)

2. Collect candidate models *(selective overview)*

Model family (examples)	Strengths	Limitations
Regression models (Linear, Ridge, Lasso)	Interpretable, simple, well understood	Limited prediction performance, few predictors
Clustering (e.g., k-means)	Identifies structure in data	Not designed for prediction tasks
Machine learning models (e.g., Neural Networks)	Strong predictive performance, flexible	Less interpretable, require tuning, can overfit

3. Select model

- **Trade-offs:** interpretability vs. performance; simplicity vs. flexibility
- **Model complexity:** flexible models capture patterns but may overfit
- **Performance:** unknown → must be **tested empirically**

4. Test and compare

→ Start with a **simple, interpretable baseline** (e.g., linear regression)
→ Then implement **more complex models for comparison**

See Provost & Fawcett (2013), chapter 3 and 4.

Regression models: A visual illustration

Model formula:

$$price = \beta_0 + \beta_1 * squarefoot$$

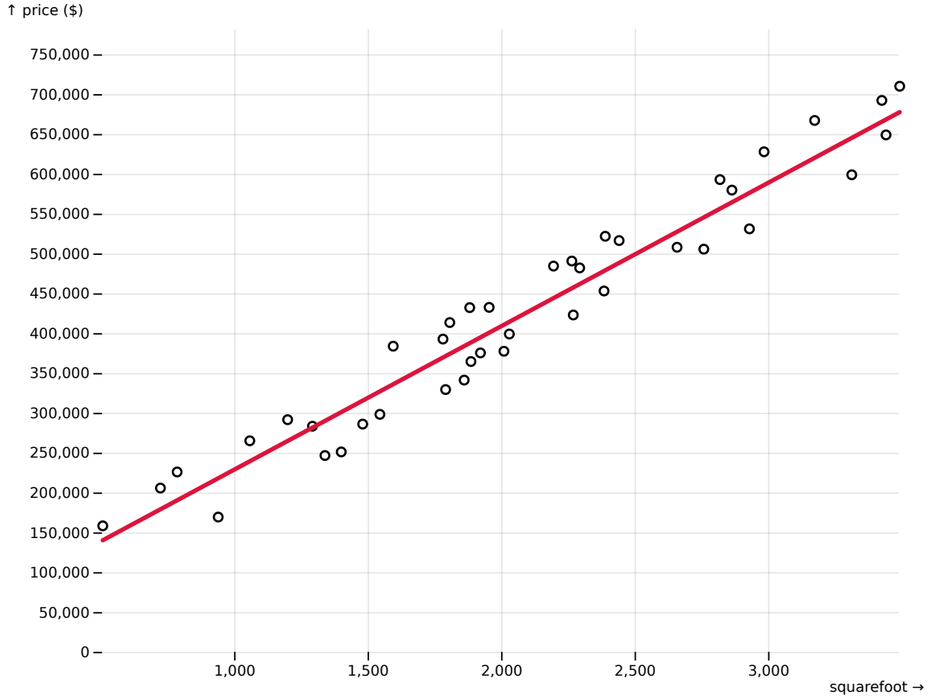
Data

squarefoot	price
3,439.185	649,728.491
1,952.616	433,264.369
2,028.285	399,838.505
721.273	206,468.719
3,490.479	710,788.526
1,879.805	432,949.25
3,171.869	667,870.811
2,382.982	453,819.253
1,920.029	376,109.648

Visualization

β_0 (base price) 500…

β_1 (price per sq ft) 180



Interpreting the regression model

The model

$$\hat{y} = \beta_0 + \beta_1 \cdot x$$

$$\hat{\text{price}} = 50{,}000 + 180 \cdot \text{squarefoot}$$

β_0 = 50,000 — *intercept*

The predicted price for a house with 0 sq ft. Rarely meaningful on its own — it anchors the line.

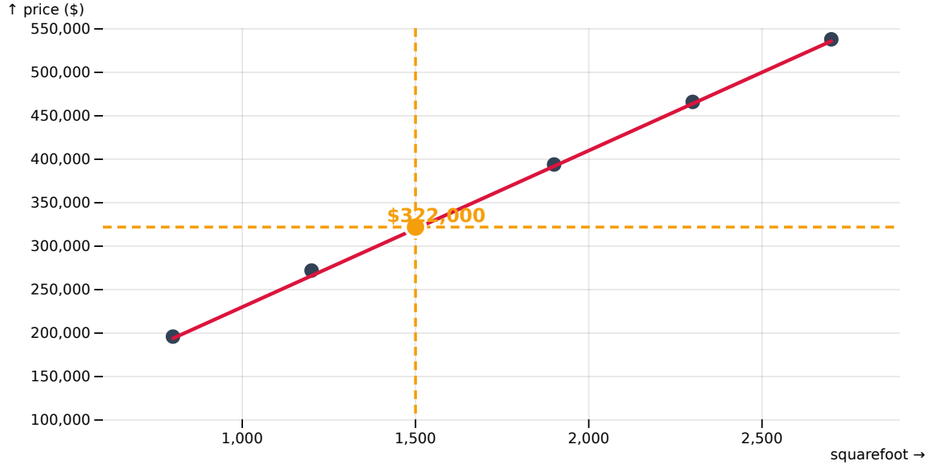
β_1 = 180 — *slope*

The **slope** (or marginal effect) of square footage: each additional square foot is associated with **+$180** in predicted price, on average.

Prediction example

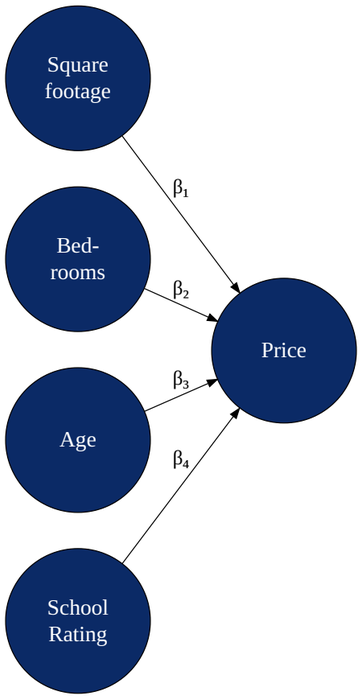
*How much would a **1,500 sq ft** house cost?*

$$\hat{price} = 50{,}000 + 180 \times 1{,}500 = \$322{,}000$$



Including multiple predictor variables

We can include many additional variables to predict the price of a house. Each coefficient (β) captures the **differential effect of a variable** —that is, how much the price is expected to change when that variable increases while the others are held constant.



As we add more predictors, the model becomes multidimensional, making it increasingly difficult to visualize.

Ordinary Least Squares Regression (OLS)

OLS is a **linear approach for predicting a quantitative response** (Y) based on a set of predictor variables X_j.

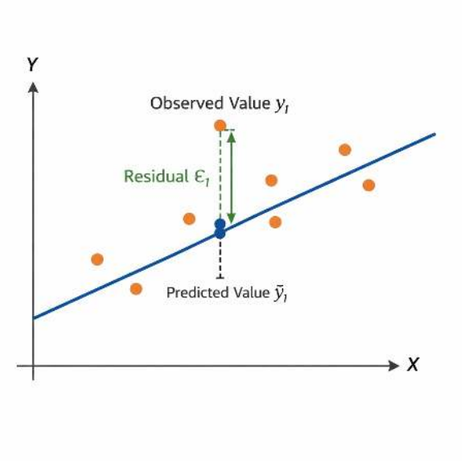
$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \cdots + \beta_p x_{p,i} + \epsilon_i$$

or in vector form

$$y_i = \beta_0 + \beta' x_i + \epsilon_i$$

The optimal **regression line** minimizes the **Residual Sum of Squares (RSS)**:

$$RSS = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta' x_i)^2$$



Matrix representation

$$y = X\beta + \epsilon$$

where

$$y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{1,1} & \dots & x_{m,1} \\ 1 & x_{1,2} & \dots & x_{m,2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1,n} & \dots & x_{m,n} \end{bmatrix}$$

Closed-form solution

The parameter vector can be estimated by

$$\beta = (X'X)^{-1}X'y$$

Learning focus

Aim to **understand and explain** the OLS procedure. You are **not required to memorize** the formulas for RSS or the closed-form OLS solution.

Implementation in Python (I)

```python
import pandas as pd
from sklearn.linear_model import LinearRegression ①

df = pd.read_csv("data/ames.csv") ②

predictors = ["squarefoot", "Overall.Qual"] ③
X = df[predictors]
y = df["price"]

model = LinearRegression() ④
model.fit(X, y) ⑤
```

① Import relevant libraries
② Load the Ames housing dataset
③ Select predictor variables
④ Create the regression model
⑤ Estimate the model on the data

Implementation in Python (II)

```python
print("Intercept:", model.intercept_)     # β0 (coefficient estimates)

coef_df = pd.DataFrame({                  # Map coefficients to variable names
    "Predictor": predictors,
    "Coefficient": model.coef_            # β1, β2, ...
})
print(coef_df)

r2 = model.score(X, y)                    # Model fit (R²)
print("R^2:", r2)
```

```
Intercept: 180,921

Predictor        Coefficient
squarefoot        110.85
Overall.Qual   28,567.43

R^2: 0.56
```

In scikit-learn, a trailing underscore indicates that an attribute is **learned from data during model fitting**. This helps to prevent accidental access to values before fitting.

- Attributes *without* _ are **hyperparameters** you specify when creating the model.
- Attributes *with* _ are **estimated after calling `.fit()`** (e.g., `coef_`, `intercept_`).

Evaluation

Evaluation

After estimating a regression model, we need to evaluate whether it is **useful and reliable**.

Evaluation focuses on three complementary questions:

1. **Can we trust the model?**
   → Are the underlying assumptions reasonably satisfied?
2. **How well does the model perform?**
   → Does it explain variation and make accurate predictions?
3. **What do the coefficients tell us?**
   → Are the estimated relationships meaningful and relevant?

Model assumptions

Regression models rely the following **assumptions**:

- Linear relationship between predictors and outcome
- Independent observations
- Constant variance of errors (*homoscedasticity*)
- Errors normally distributed

After fitting a model, we can evaluate whether these assumptions are reasonable. Different violations affect different aspects of the model (interpretation, uncertainty, prediction).

Assumption violated	**Coefficients** *(interpretation)*	**Confidence intervals / p-values** *(uncertainty)*	**Prediction** *(performance)*
Linearity	❌ biased	invalid	⚠️ worse
Independence	✅ OK	too optimistic	⚠️ context-dependent
Homoscedasticity	✅ OK	incorrect SEs	✅ mostly OK
Normality	OK	⚠️ small-sample issue	✅ OK

💡 The impact of assumption violations depends on the **characteristics of the dataset** (e.g., sample size, noise, structure) and the **goal of the analysis**.

Overall model

A common measure to assess the performance of a regression model is R^2 (the coefficient of determination):

$$R^2 = 1 - \frac{RSS}{TSS}$$

- Measures the **share of variance in the target variable explained by the model**
- Example: **(R^2 = 0.56)** means the model explains **56% of the variation in house prices**
- Values range from **0 to 1**, with **higher** R^2 generally indicating a better fit
- But a high R^2 **does not automatically mean** the model is useful in practice. R^2 says little about:
  - whether predictions are accurate enough for decisions
  - whether the model generalizes to new data
  - whether coefficients should be interpreted causally

Other evaluation measures

- **MAE (Mean Absolute Error)** Average absolute prediction error → easy to interpret in the unit of the target variable
- **RMSE (Root Mean Squared Error)** Penalizes large errors more strongly → useful when large mistakes are especially costly

Coefficients

Regression coefficients tell us how the predicted outcome changes when a predictor changes, **holding the other predictors constant**.

For a coefficient β_j:

- **Sign** Positive: higher x_j is associated with higher predicted price Negative: higher x_j is associated with lower predicted price
- **Magnitude** Indicates the expected change in the target variable for a one-unit increase in x_j

Example:

- $\beta_{\text{squarefoot}}$ = 110.85 → one additional square foot is associated with about **+$110.85** in predicted price
- $\beta_{\text{Overall.Qual}}$ = 28,567.43 → one additional quality point is associated with about **+$28,567** in predicted price

When evaluating coefficients, consider both:

1. Statistical significance

- Asks whether the estimated relationship is likely different from zero
- Commonly assessed with **standard errors**, **t-tests**, **p-values**, **confidence intervals**

2. Practical significance

- Asks whether the effect is **large enough to matter in practice**
- A coefficient can be statistically significant but still have little managerial or business relevance

Next steps

Next steps

Once we move beyond a simple regression model, practical follow-up questions arise:

- **How can we use categorical predictors?**

  → See next slides.

- **Which predictors should we include?**

  → Approaches such as **forward selection** and **backward selection** help identify a useful subset of variables
  → **Forward selection** starts with a very simple model and **adds predictors step by step** if they improve the model
  → **Backward selection** starts with a model containing many predictors and **removes the least useful ones step by step**

- **Why not include all available variables?**

  → Adding too many predictors can create problems such as:

  - **dependence between predictors**: predictors overlap strongly, making coefficients unstable
  - **overfitting**: the model fits the training data well but performs poorly on new data

> **Takeaway**
>
> A good regression model is usually not the one with the **most variables**, but the one that achieves a good balance between **interpretability**, **stability**, and **predictive performance**.

Categorical variables

Regression models require **numerical input**.

But real-world datasets often include **categorical variables**, for example:

- Industry: *Finance, Healthcare, Retail*
- Region: *EU, US, APAC*

Question:
How can we include such variables in a regression model for `performance`?

Categorical variables: A naïve solution

We could assign numbers:

```
Finance = 1, Healthcare = 2, Retail = 3
```

Problem:

- This introduces an **artificial order**
- The model assumes:
  - Retail > Healthcare > Finance

Regression model:

$$\text{performance}_i = \beta_0 + \beta_1 \cdot \text{Industry}_i + \varepsilon_i$$

Encoding:

$$\text{Finance} = 1,\ \text{Healthcare} = 2,\ \text{Retail} = 3$$

Implications

- **Ordering:** Retail > Healthcare > Finance
- **Equal marginal effects:**

$$\text{Effect}(1 \to 2) = \text{Effect}(2 \to 3) = \beta_1$$

Key issue

Categorical variables have **no natural order or distance**, but the model treats them as **equally spaced numeric values**. We must avoid introducing **false relationships**.

Categorical variables: One-hot encoding

Create **one binary column per category** (1: category present, 0: category not present).

Example transformation:

Original data

Industry	Performance
Finance	120
Retail	95
Healthcare	140

After one-hot encoding

Performance	Retail	Healthcare
120	0	0
95	1	0
140	0	1

Advantages

- No artificial ordering
- The model learns **separate effects** (one β_i coefficient for each category)

One-hot encoding allows categorical variables to be used in linear regression models.

Notice that `Finance` is dropped and serves as the **reference category**. The remaining dummy coefficients show differences relative to `Finance`.

Categorical variables: Interpreting coefficients

Assume the following regression model:

$$\text{Performance}_i = 100 + 20 \cdot \text{Retail}_i + 40 \cdot \text{Healthcare}_i + \varepsilon_i$$

Interpretation (reference: Finance)

- Intercept (100): baseline performance for **Finance**
- Retail (20): **+20** relative to Finance → 120
- Healthcare (40): **+40** relative to Finance → 140

Key idea

Coefficients measure **differences relative to the reference category**.

Deployment

From model to use in practice

Once a regression model has been estimated and evaluated, it can be **deployed** to support decisions in practice.

Typical uses include:

- **Describe** Identify which factors are associated with higher or lower prices
- **Predict** Estimate the expected price for a new house based on its characteristics
- **Prescribe** Use predictions as input for action, for example:
  - setting an asking price
  - prioritizing properties for review
  - comparing renovation options

> **Key idea**
>
> A regression model does not only help us **understand** the data. It can also be embedded in workflows to support **future decisions**.

Deployment example: Making a prediction

Suppose our fitted model is:

$$\widehat{\text{price}} = 180{,}921 + 110.85 \cdot \text{squarefoot} + 28{,}567.43 \cdot \text{Overall.Qual}$$

For a house with:

- `squarefoot = 1500`
- `Overall.Qual = 7`

the predicted price is:

$$\widehat{\text{price}} = 180{,}921 + 110.85 \cdot 1500 + 28{,}567.43 \cdot 7$$

$$\widehat{\text{price}} \approx 547{,}150$$

Deployment example in Python

A fitted model can be **saved** and **loaded** for later use.

```python
import joblib

# Save trained model
joblib.dump(model, "house_price_model.joblib")

# Load trained model later
loaded_model = joblib.load("house_price_model.joblib")

# New data for prediction
new_house = pd.DataFrame([{
    "squarefoot": 1500,
    "Overall.Qual": 7
}])

predicted_price = loaded_model.predict(new_house)
```

Why save the model?

Saving a model makes it possible to reuse it in applications, dashboards, scripts, or decision-support systems without fitting it again each time.

Deployment considerations

Before using a regression model in practice, we should ask:

- **Does it generalize?** Does it still perform well on new data?
- **Is it robust?** Are predictions stable when conditions change?
- **Is it interpretable?** Can decision makers understand how outputs are generated?
- **Is it used responsibly?** Could predictions reinforce bias or lead to unfair decisions?

> **Takeaway**
>
> Deployment is not the end of the analytics process. Models should be **monitored, reviewed, and updated** as data, environments, and decision needs change.

Vocabulary

An **algorithm** is a procedure or set of steps or rules to accomplish a task. It is usually the implementation of a method. Algorithms are used to build models.

In the given context, a **model** is the description of the relationship between variables. It is used to create output data from given input data, for example to make predictions.

A **predictor** is a variable used as an input to a model to explain or predict an outcome. Synonyms include: Independent variable (IV); explanatory variable; regressor (econometrics); feature (machine learning); input variable; covariate (statistics, esp. causal inference); control variable (when included to adjust for effects); factor (sometimes, esp. experimental settings); attribute (data mining).

An **outcome** refers to the variable a model aims to explain or predict based on the predictors. Synonyms include: Dependent variable (DV); response variable; target; label (classification); explained variable; criterion variable; output variable.

Fitting a model means that you estimate the model using the observed data. You are using your data as evidence to help approximate the real-world mathematical process that generated the data. Fitting the model often involves optimization methods and algorithms, such as maximum likelihood estimation, to help get the parameters.

Overfitting arises when a model has learned sample-specific patterns (including noise) rather than the true data-generating process, leading to low out-of-sample predictive performance.

— Source: Schutt & O'Neil (2014)

Summary

- Regression is a core example of a structured, model-based analytics process: from business question to data, modeling, evaluation, and potential deployment.
- Linear regression models quantify relationships between variables through coefficients, estimated via optimization (OLS). Coefficients capture **marginal effects** (numeric variables) and **differences relative to a reference category** (categorical variables using one-hot encoding).
- In practice, the analytics workflow can be implemented in Python: define a model, fit it to data, generate predictions, and evaluate performance — a pattern that extends to many other modeling approaches.

Survey: Session 4

Please complete the survey before you leave today — thank you 🙏

https://forms.gle/GtBNSMdCTZ92ix549

Note: Responses may be analyzed and published in anonymized form.

References

Provost, F., & Fawcett, T. (2013). *Data science for business: What you need to know about data mining and data-analytic thinking*. " O'Reilly Media, Inc.".

Schutt, R., & O'Neil, C. (2014). *Doing data science*. O'Reilly Media. https://www.oreilly.com/library/view/doing-data-science/9781449363871/